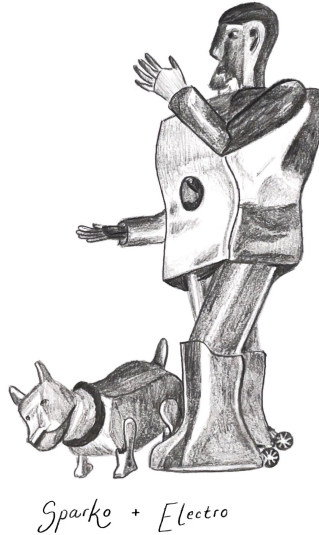


Will Robots Take Over the World? The Effects of AI on Jobs

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1 Introduction

When you think of the future, you may envision a world run by robots, where humans can relax and sit back while an android does your chores for you. Studies show that this may become reality sooner than you may expect.

2 The Past: The History of Robots and AI

2.1 The Beginning

Artificial Intelligence is defined as “the ability of a computer or a robot controlled by a computer to do tasks that are usually done by humans.” The idea of robots was first presented in 1920 by Karel Capek, in his play *Rossum’s Universal Robots*. The play presented robots taking over the world, giving birth to the science fiction genre.

The first robots were displayed in 1939 at the World’s Fair by the Westinghouse Electric Company. “Elektro the Moto-Man” had the ability to walk, talk, smoke cigarettes, blow up balloons, and move its head and arms. Elektro appeared again in the 1940 World’s Fair, along with Sparko, a robot dog who could sit, bark, and beg.

Following the revolutionary creations of Elektro and Sparko, Isaac Asimov, a professor at Boston University, created the three laws of robotics:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given to it by human beings except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

These laws showed that robotics was beginning to be recognized as a prominent field of science.

2.2 Artificial Intelligence



Figure 1: W. Gray Walter and one of his robots

While Elektro and Sparko were advanced for their time, they required voice commands to function. The first official artificial intelligence came in the form of two robot tortoises, Elmer and Elsie. Built by William Grey Walter in the late 1940s, these autonomous robots utilized sensory inputs by perceiving light and feeling with a bump sensor. These advancements developed not only technology but also our understanding of the human nervous system.

3 The Present: Modern Advancements

3.1 Horses

Many people question whether robots can take over humans' jobs. However, machines have already made one work animal obsolete: horses.

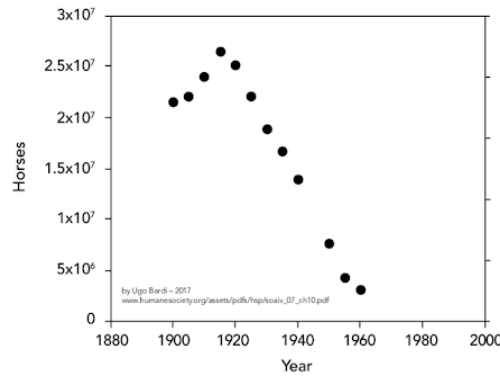


Figure 2: Horse population over time

As the graph depicts, the population of horses peaked in 1915 but sharply declined soon after. This is directly linked to the fact that cars began to be commonly used in the 1920s. As cars became more available, horses became unnecessary for human use. Other roles traditionally held by horses were also impacted. Technology started to replace horses in transportation, agriculture, warfare, and more.

3.2 Current Robots/AI Replacing Jobs



While there may be doubts about how quickly robotic replacement can happen with humans, there are already intelligent robots among us. An example of this is Baxter, a robot with vision who can “learn what you want him to do by watching you do it.” Baxter’s skill set includes industrial tasks such as “loading, unloading, sorting, and handling of materials.” While the robot takes much more time performing tasks when compared to a human, it is also much more cost-effective. Baxter costs about \$25,000 to create and maintain, which is about \$40,000 less than the average household income in America.

While this may sound like a complex idea, there are already examples of artificial intelligence replacing human jobs. In new supermarkets, robotic cashiers have replaced humans; “what used to be 30 humans is now 1 human overseeing 30 cashier robots.” Another example is the surge in robot baristas; over 500,000 Americans work as baristas, and their jobs are in danger of being replaced by artificial intelligence.

Many other fields are also impacted by robots. Chatbots now lead customer service and can help users do anything, from opening an account to transferring money. In addition, self-driving cars are emerging. The transportation industry currently consists of at least 70 million workers worldwide. Self-driving cars will likely replace these human drivers just as cars replaced horses. Ultimately, robots' ability to think quickly with low chance of error allow them to take human jobs.

3.3 Professional Bots

Careers such as lawyers and doctors require years of education and hours of dedicated work every day. However, lawyers and doctors are still humans, and in the words of Alexander Pope, "to err is human." For example, lawyers must look through countless legal documents to find details supporting their case. Missing just one paragraph can severely increase their workload. Professional bots can look through millions of emails or documents in a matter of hours, whereas it would take humans weeks. The bots outweigh humans in cost efficiency, time, and accuracy. All of these factors lead to a boost in the economy as well, since more work is accomplished in the same amount of time.

Artificial intelligence can also leave a mark on the medical industry. Humans are not capable enough to understand all medicines and their interactions with each other. By analyzing symptoms and comparing them to a database, robots can accurately diagnose patients and prescribe them the necessary drugs. While it is impossible for a human to know how every medicine affects the human body, this is not an issue for robots. Bots can also use the experience of other robots to improve themselves.

Watson, a robot created by IBM, is providing guidance for lung cancer treatments at Memorial Sloan Kettering Medical Center. Watson is an example of how artificial intelligence can perform human tasks more efficiently. Watson has also made a guest appearance on 'Jeopardy!', where it showcased its ability to quickly solve a variety of trivia questions.



While lawyers and doctors will most likely stay, their need and numbers will decrease in the near future.

3.4 Self-Teaching Bots

Breakthroughs in technology have even reached self-teaching robots. This may sound preposterous, but these bots are essential to America's economy. The stock market, which used to be run by humans, is now run by "bots who taught themselves to trade stocks trading stocks" with other

such bots. This is a matter of concern to human workers, as it shows the capability of artificial intelligence to learn and replace most, if not all, jobs.

4 The Future: Paradise or Dystopia?

4.1 Long Term Effects of AI on Jobs

Robots already play a significant role in our economy. As they begin to take more jobs from people, humanity will be drastically impacted. It's not wrong for people to believe that technology will make life easier and allow more leisure for humans. However, convenience comes with a cost. In the same way that technology took away horse labor, AI can have detrimental effects on human labor.

There are many debates about whether or not AI will replace all jobs. Some argue that bots can teach themselves to do any work, while others say that careers which involve emotions, such as therapists, are safe. Nevertheless, most agree that robots can and will replace most jobs. Since AI is more efficient than humans, the economy will likely grow, while the income of humans will decrease.

4.2 A Proposed Solution

The coming wave of AI is undeniable. The only action humanity can take is to prepare for it. Some politicians agree that the best way to help the citizens is a Universal Basic Income, or UBI, where the government pays a set amount of money to all citizens. This might help families sustain themselves even when their job is replaced by AI. Others are opposed to this idea, believing that it will make people underproductive and overdependent on the government.

5 Conclusion

Robots will continue to replace humans in the work world; it is not a matter of if, but rather, when. Politicians and scientists must work together and find a way to help people with diminished income or obsolete jobs. If they don't, people may find their own creations working against them.



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